# ARUN KUMAR ARASAPPAN

**Senior Backend Engineer | Python, Django & Cloud Technologies**

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## PROFILE SUMMARY

Senior Backend Engineer with 8 years of experience designing, building, and supporting scalable Python and Django applications across intelligent document processing, enterprise automation, and analytics platforms. Experienced in developing cloud-native backend services, REST APIs, distributed systems, CI/CD pipelines, and observability solutions on AWS and Azure, delivering measurable improvements in application reliability, release efficiency, and platform performance.

Hands-on experience integrating LLMs, BERT-based NLP, Retrieval-Augmented Generation (RAG), and agentic AI into production-grade document intelligence solutions. Recognised for technical ownership, clean architecture, cross-functional collaboration, and delivering secure, maintainable software that supports evolving business and customer needs.

## SKILL SET

- **Programming Languages:** Python | Go (Golang) | SQL | JavaScript | Bash
- **Backend Development:** Django | Django REST Framework (DRF) | RESTful API Development | API Design | Celery | Asynchronous Processing
- **Frontend:** React
- **Databases:** PostgreSQL
- **AI & Intelligent Automation:** Large Language Models (LLMs) | Intelligent Document Processing (IDP) | Retrieval-Augmented Generation (RAG) | LangChain | Agentic AI | Model Context Protocol (MCP) | PyTorch | BERT (NER)
- **Cloud, DevOps & Infrastructure:** AWS | Microsoft Azure | Docker | Terraform | GitHub Actions | Azure DevOps | Jenkins | CI/CD | Infrastructure as Code (IaC)
- **Observability & Monitoring:** Prometheus | Grafana | ELK Stack | Application Monitoring | Centralised Logging
- **Software Engineering & Architecture:** Backend Architecture | Distributed Systems | System Design | Scalable Application Development | Performance Optimisation | Caching
- **Security & Development Practices:** Authentication & Authorisation | Role-Based Access Control (RBAC) | Agile Development | Technical Mentoring

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer | Xerox, India | June 2023 – Present

- Lead backend engineering for an enterprise Intelligent Document Processing (IDP) platform using Python and Django REST Framework.
- Design scalable REST APIs, distributed processing pipelines, and backend architecture supporting enterprise document ingestion and extraction.
- Collaborate with cross-functional engineering teams to drive architecture decisions, release planning, and technical best practices.

***Key Achievements***

- Increased document extraction throughput and reduced processing errors by 20% through scalable backend architecture enhancements.
- Integrated BERT-based NER, LLM capabilities, and RAG workflows to improve extraction accuracy for unstructured enterprise documents.
- Reduced production recovery time by 35% by implementing Prometheus, Grafana, and ELK-based observability solutions.
- Shortened QA validation cycles by 40% through automated API regression testing and pre-release verification workflows.
- Standardised CI/CD pipelines using GitHub Actions, Azure DevOps, Docker, and Terraform while mentoring engineers on backend engineering practices.

### Full Stack Developer (Django) | Neo Horizon Analytics, India | April 2021 – May 2023

- Developed full-stack enterprise applications using Django, Django REST Framework, React, and PostgreSQL for analytics-driven business solutions.
- Designed secure REST APIs, scalable backend services, and responsive user interfaces supporting high-volume data processing and visualisation.
- Partnered with business stakeholders to translate functional requirements into scalable, maintainable, and production-ready software solutions.

***Key Achievements***

- Improved application responsiveness by 30% through SQL optimisation, Django ORM tuning, and front-end performance enhancements.
- Delivered a Django-based Warehouse Management System that standardised inventory operations and reduced manual data-entry errors.
- Strengthened application security by implementing token authentication, role-based access control, API versioning, and request throttling.
- Enhanced application scalability by introducing Celery background processing for automated reporting and asynchronous data export workflows.

### Software Engineer | ThinkPalm Technologies, India | June 2019 – April 2021

- Delivered Python-driven automation solutions that improved engineering productivity and streamlined software delivery across development teams.
- Supported cross-functional product delivery by translating technical requirements into scalable automation and reporting solutions.

***Key Achievements***

- Engineered reusable Python automation frameworks, eliminating repetitive engineering activities and reducing manual effort by approximately 50%.
- Created interactive Python and React dashboards, providing engineering and quality insights that strengthened release planning and delivery decisions.
- Facilitated collaboration across engineering teams to resolve technical issues, refine requirements, and improve feature delivery efficiency.

### Associate Test Engineer (Contract) | HCL Technologies, India | Sep 2018 – May 2019

- Contributed to software quality assurance by supporting Python-based automation testing for enterprise print-driver applications.

***Key Achievements***

- Transformed manual regression test scenarios into reusable Python automation scripts, improving testing efficiency and release readiness.
- Embedded automated regression execution within software release processes while maintaining defect traceability through IBM Rational ClearQuest.

## EDUCATION

**Bachelor of Engineering** - **Mechatronics Engineering** | Mahendra Engineering College, India | April 2018
**(GPA: 7.5 / 10)**